



Customer Retention & Churn Analysis Dashboard

Future Interns – Data Science & Analytics

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Task: Customer Retention & Churn Analysis

Tools Used

- Microsoft Power BI
- Power Query
- DAX
- Microsoft Excel

Dataset: Telco Customer Churn Dataset

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Introduction

Customer retention is one of the most important success factors for subscription-based businesses. Acquiring new customers is generally more expensive than retaining existing ones. Therefore, analysing customer churn helps organisations understand customer behaviour, identify reasons for customer loss, and improve long-term business performance.

This project focuses on analysing customer retention using the Telco Customer Churn dataset. An interactive Power BI dashboard has been developed to visualise churn patterns, customer segments, payment methods, contract types, and customer lifetime metrics. The analysis aims to provide actionable insights that can help improve customer retention strategies.

Objectives

The main objectives of this project are:

- Analyse customer churn patterns.
- Measure customer retention rate.
- Identify customer segments with high churn.
- Analyse contract type and payment method.
- Study customer tenure behaviour.
- Build an interactive Power BI dashboard.
- Generate business recommendations.

Dataset Description

- Dataset Information

Attribute	Value
Dataset Name	Telco Customer Churn
Source	Kaggle
Records	7043
Columns	21
File Format	CSV

- Important Features

Column	Description
customerID	Customer Identifier
gender	Gender
SeniorCitizen	Senior Citizen Status
tenure	Customer Lifetime (Months)
Contract	Contract Type
InternetService	Internet Service
PaymentMethod	Payment Method
MonthlyCharges	Monthly Charges
TotalCharges	Total Charges
Churn	Customer Status

Data Cleaning & Dashboard Development

○ Data Cleaning

The following preprocessing steps were performed before analysis:

- Imported CSV dataset into Power BI.
- Verified data types.
- Removed blank values from TotalCharges.
- Checked missing values.
- Validated column consistency.

○ DAX Measures Created

- Total Customers
- Active Customers
- Churn Customers
- Churn Rate
- Retention Rate
- Average Monthly Charges
- Average Tenure

Dashboard Components

KPI Cards

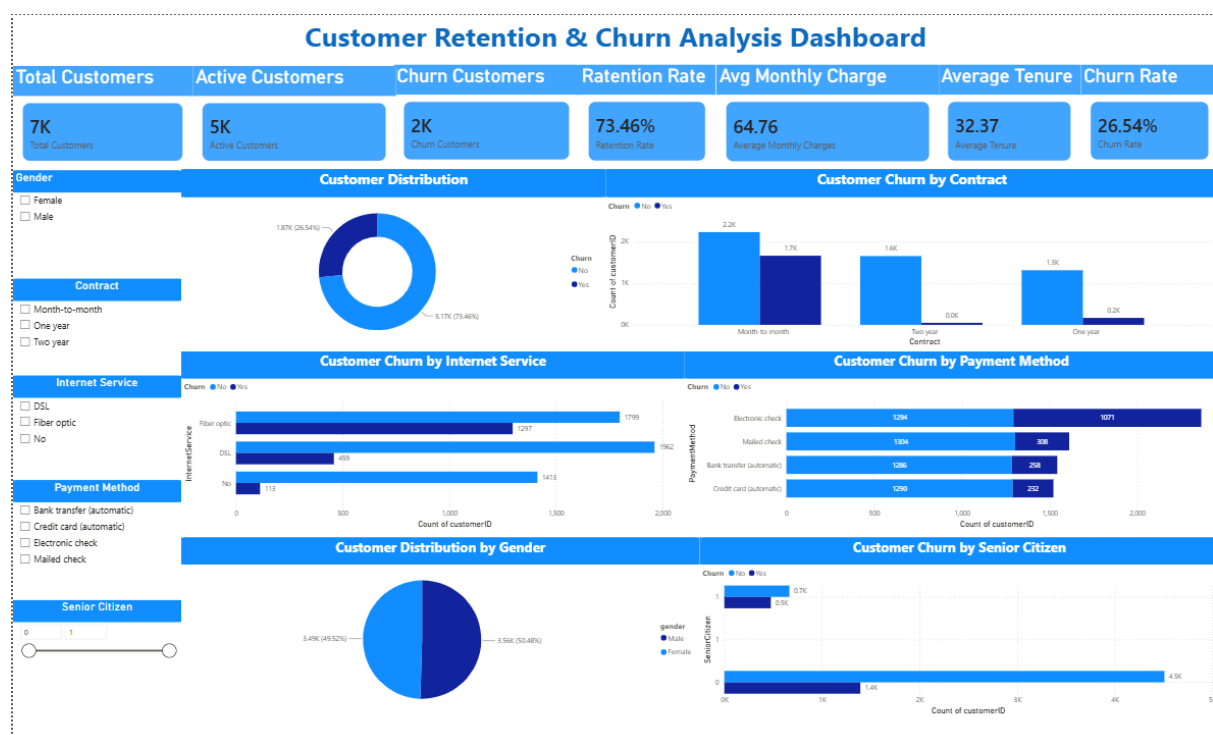
- Total Customers
- Active Customers
- Churn Customers
- Churn Rate
- Retention Rate
- Average Monthly Charges
- Average Tenure

Visualisations

- Customer Distribution
- Churn by Contract
- Churn by Internet Service
- Churn by Payment Method
- Gender Distribution
- Senior Citizen Analysis

Filters

- Gender
- Contract
- Internet Service
- Payment Method
- Senior Citizen



Business Insights

Key Insights

1. Overall Churn

The overall customer churn rate is **26.54%**, indicating that approximately one out of every four customers has left the company.

2. Contract Type

Customers with Month-to-Month contracts show the highest churn, whereas Two-Year contracts have the strongest customer retention.

3. Internet Service

Customers using Fiber Optic internet services experience higher churn than DSL users.

4. Payment Method

Electronic Check users demonstrate a higher churn rate than customers using automatic payment methods.

5. Customer Tenure

Customers with longer tenure are significantly more likely to remain with the company.

6. Gender

Customer distribution is nearly balanced between male and female customers, with no significant difference in churn.

7. Senior Citizens

Senior Citizen customers exhibit a relatively higher churn rate than non-senior customers.

Business Recommendations

Recommendations

1. Encourage Month-to-Month customers to upgrade to yearly contracts through discounts and loyalty benefits.
2. Improve the quality and support of Fiber Optic internet services.
3. Promote automatic payment methods by offering cashback or reward programmes.
4. Launch retention campaigns for customers with low tenure.
5. Develop personalised offers for customers identified as high churn risk.
6. Provide dedicated customer support and tailored service plans for Senior Citizens.

Conclusion & References

Conclusion

This project successfully analysed customer retention and churn behaviour using Power BI. The interactive dashboard highlighted key churn patterns based on contract type, internet service, payment method, and customer demographics. The findings demonstrate how data analytics can support business decision-making and improve customer retention through targeted strategies.

References

- Kaggle – Telco Customer Churn Dataset
- Microsoft Power BI Documentation
- Microsoft Learn
- Future Interns – Data Science & Analytics Task Guidelines